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# Influence of empathic concern on fairness-related decision making: Evidence from ERP

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**Abstract**

Using event-related potential (ERP) and ultimatum game (UG), this study investigated the influence of empathic concern on fairness-related decision making. The experiment adopted a 2 (state empathic concern: empathy vs. non-empathy) × 3 (fairness: fair vs. disadvantageous unfair vs. advantageous unfair) within-subject design. A total of 37 participants participated in the experiment, and they were asked to choose whether to accept offers from different proposers as responders. Behavior results showed that the acceptance rate of empathy condition was higher than that of non-empathy condition for disadvantageous unfair offers, and the opposite result was observed for advantageous unfair offers. ERP results revealed that for disadvantageous unfair offers, the non-empathy condition elicited a more negative-going anterior N1 (AN1) than the empathy condition, and the empathy condition elicited a larger P2 amplitude than the non-empathy condition. In the empathy context, the disadvantageous unfair condition elicited more negative-going medial frontal negativity (MFN) than the advantageous unfair and fair condition. P3 of fair condition was larger than that of disadvantageous unfair condition, which was not modulated by empathy. These results indicated that empathy modulated not only fairness-related decision making behavior, but also early attention and motivation as well as later cognitive and emotional processing in fairness. However, the higher cognitive processes characterized by P3 were only modulated by fairness but not affected by empathy.

**Key words** fairness-related decision making, empathic concern, AN1, P2, MFN

## 1 Introduction

Empathy, as an ability to deeply feel and understand other people's emotions (Decety & Lamm, 2006), has received extensive attention and research. With the deepening of research, researchers put forward that empathy can have an impact on decision-making, and this view has been scientifically verified from questionnaire survey, behavioral experiments and electrophysiological research. According to the empathy-altruism hypothesis, when other people are in distress, an individual develops a range of emotions including empathic concern, sympathy, compassion and an altruistic motive to relieve distress, leading to altruistic behavior (Batson, 1991; Batson & Shaw, 1991). Therefore, researchers have focused on the influence of empathy on altruistic decision-making (FeldmanHall et al., 2015; McAuliffe et al., 2018; Liu et al., 2018, 2020). The new research gradually shifts the research perspective to other decision-making types, such as fairness-related decision making that is closely related to social life: It was found that empathic concern toward the responder increased third-party punishment behaviors at the cost of self-interest in unfair situations when the individual observed the inequity as a third party (Leliveld et al., 2012; Pfattheicher et al., 2019). Previous studies have mostly focused on the positive role of empathy in promoting fairness-related decision making. However, empathy will also interfere with decision-making in certain situations. For example, when we empathize with the perpetrators of unfair behaviors, we are more likely to accept their unfair behaviors. Then, when an individual empathizes with unfair implementers, how does empathy weaken the fairness motivation in the corresponding situation? For one thing, exploring this issue can provide richer evidence to support the influence of empathy on decision making from different perspectives, which helps us understand the advantages and disadvantages of empathy more comprehensively. For another, it can also help us to understand some social phenomena, such as empathy for the perpetrators can affect the public's fair judgment, and even some media utilize it to manipulate public opinion to interfere with the justice. Therefore, we hope to explain these phenomena from the perspective of psychological cognitive process and neural mechanism, and apply them to public education and public opinion supervision, so as to urge everyone to treat social events more rationally and fairly. In this study, we want to simulate a real situation to induce empathic concern state to investigate the influence of empathy toward unfair implementer on fairness-related decision making process related to individuals' interests, as well as to understand the Temporal dynamics characteristics of neural activities.

### 1.1 Definition and paradigm of fairness-related decision making

With the society and the country constantly emphasizing fairness and justice, fairness norms play an increasingly critical role in various social and economic decisions. Fairness-related decision making refers to following the principle of fairness when making decisions to obtain benefits and assume responsibilities (Qi et al., 2017). Ultimatum game (UG) paradigm is a

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classic paradigm for studying fairness-related decision making, which consists of two players: the proposer needs to make an offer to allocate a certain amount of money, and the responder chooses to accept it or not. If they accept it, the offer will be allocated according to that offer. If they reject it, neither participant will get any benefit (Güth et al., 1982). Studies using UG have found that individuals generally show inequity aversion in decision making (Fehr & Schmidt, 1999), which is reflected in negative emotions caused by unfairness and the rejecting behaviors at the cost of their own interests (Camerer, 2003; Polezzi et al., 2008; Sanfey et al., 2003), reflecting individuals' concern about fairness.

### 1.2 Influence of empathic concern on fairness-related decision making

In fact, the fairness-related decision making process could be affected by various factors, and what is finally shown is not absolute equality, but relative and slightly biased fairness. These factors cover individual, situational and social aspects, including social value orientation, framing effect, social distance and group identity (Bieleke et al., 2017; Wang et al., 2014; Wu & Zhou, 2012). Among them, empathy is a quite important factor affecting fairness processing, which is closely related to our life. For example, in the process of resource allocation, if others are in need, then we are faced with the need to maintain fairness, as well as the need to care for others and safeguard their interests. In this situation, do we still focus only on fairness? Or will it be influenced by empathy? Empathic concern (also known as empathic care or compassion) is an emotional and motivational state characterized by helping and promoting the welfare of others (Bernhardt & Singer, 2012; Light et al., 2015). Specifically, when people observe or imagine that others are facing misfortune and suffering, they will develop a state of empathic concern and trigger a series of emotional reactions toward others, including sympathy and compassion, which motivate individuals to reduce the suffering of others (Batson et al., 2007). Studies have explored the effects of empathy on fairness-related decision making in situations unrelated to self-interest (Leliveld et al., 2012; Pfattheicher et al., 2019), but no research has paid attention to how empathy would affect fairness processing when individuals are direct stakeholders (directly encountering unfairness). Specifically, a conflict situation may be formed when considering empathy: empathic concern induces altruistic motivations that motivate individuals to care for others (Batson et al., 2007), while individuals are also concerned with fairness, even at the cost of self-interest to punish unfairness (Camerer, 2003; Fehr & Gachter, 2002). When these two goals or motivations are in conflict, how does the individual deal with the relationship between them, i.e., does the individual prefer altruism or fairness in response to an unfair offer from another person in need? This question can help us understand the behavior patterns of individuals in specific situations.

### 1.3 Time course characteristics of fairness processing

Based on the discussion above, we hypothesized that empathic concern would affect individuals' fairness-related decision making behavior. To further confirm which stages of fairness processing were influenced by empathy, we used event-related potential (ERP) technique with high sensitivity and temporal resolution, hoping to reveal the fairness processing influenced by empathy from the temporal dimension. According to previous studies, ERP components related to fairness processing include anterior N1 (AN1), P2, medial frontal negativity (MFN) and P3 (Boksem & De Cremer, 2010; Hu & Mai, 2021; Wang et al., 2014).

Both AN1 and P2 are early components, reflecting early automated processing (Eisenberger, 2015; Horat et al., 2016). AN1 is a negative component that appears approximately 100 ms after visual stimulus onset and peaks around 120 ms, distributed in the central frontal region (Boudreau et al., 2009). AN1 is associated with early automatic attention, with attended stimuli eliciting larger AN1 than non-attended stimuli (Duzcu et al., 2019). Because unexpected stimuli usually attract attention more easily, unexpected stimuli also elicit larger AN1 amplitude (Annic et al., 2014; Bouwer et al., 2020). The effects of AN1 in social decision-making studies have also been paid attention to: conflict of interest induces more negative-going AN1 than congruent interest decision-making situations (Boudreau et al., 2009). In the study of fairness-related decision making, group identity can modulate AN1 amplitude by modulating the individual's expectation of the offers from in-group and out-group members, with greater AN1 amplitude reflecting expectation violations of different offers (Wang et al., 2014). To sum up, it is clear that AN1 can be modulated by stimulus characteristics and the object of interaction. P2 is a positive component that peaks around 200 ms, usually distributed in the frontal and central regions (Horat et al., 2016). Like AN1, P2 is also associated with early attention allocation (Luck & Hillyard, 1994), except that P2 is also modulated by motivational relevance or emotive meaning. Stimuli with stronger motivation prominence can prompt individuals to allocate more attentional resources to them, thus eliciting larger P2 amplitude (Carretie et al., 2004). In interpersonal decision-making, interest congruent situations are more motivating and perceptually salient for individuals, and thus induce greater P2 amplitude than conflict of interest situations (Boudreau et al., 2009). Studies using UG have found that fair offers induced a larger P2 amplitude than unfair offers, reflecting individuals' intrinsic goals and motivations to pursue fairness (Hu & Mai, 2021; Wu et al., 2011). In previous studies, AN1 and P2 effects were mainly reflected in the processing of pain empathy, reflecting the early effect of pain scene response and the automatic processing of pain empathy, which were important indicators of early automatic emotional activation and sharing mechanisms (Decety et al., 2010; Fan & Han, 2008). Although no studies have identified AN1 and P2 effects of empathy in social contexts, numerous studies have found that physical and social pain share underlying neurophysiological mechanisms, including the dorsal anterior cingulate and right ventral prefrontal cortex (DeWall et al., 2010; Eisenberger, 2015; Eisenberger et al., 2003; Kross et al., 2011). Besides, experiencing unfairness can cause individuals to experience social pain (Campanha et al., 2011). Therefore, given the similar processing patterns of physical pain and social pain, we hypothesized that empathy would also influence AN1 and P2 effects during fairness processing in social contexts.

MFN is a negative component distributed in the central frontal, peaks within 250~450 ms after stimulus onset, reflecting rapid semi-automatic processing of the current stimulus, indicating the middle stage of fairness processing (Leng & Zhou, 2010). MFN is thought to be related to negative outcomes (Boksem & de Cremer, 2010), violation of social norms and expectation deviation (Hewig et al., 2011; Van der Veen & Sahibdin, 2011; Wu et al., 2011). In UG, whether advantageous unfair or disadvantageous unfair offers elicited a more negative-

going MFN than fair offers (Boksem & de Cremer, 2010; Hewig et al., 2011; Spape et al., 2019). Meanwhile, MFN also reflected affective appraisal of negative outcomes, since unfair offers were usually thought to be worse, which would lead to negative emotional reactions (Moser et al., 2014). Importantly, MFN could be modulated by empathy, as Liu et al. (2018, 2020) investigated the influence of empathic concern on outcome evaluation. It was found that the neural activity elicited by self-gain and self-loss was modulated by empathy: in the situation without conflicting interests, MFN effect was significantly larger when participants were induced with empathy state than when they were not. Also, the MFN responses to stranger's outcomes in empathy condition were as strong as those to themselves. In the situation of conflicting interests, winning money for others of non-empathy condition means their own losses, which elicited greater MFN, while the reversed MFN pattern was not observed in the empathy condition, suggesting that the neural responses to one's own loss were buffered by empathy.

The late stage of fairness processing is represented by P3, which is a centro-parietal positivity appearing in the 300~600 ms time window. It is considered to be closely related to advanced cognitive processes such as selective attentional allocation and high-level affective evaluation (Hu & Mai, 2021; Leng & Zhou, 2010; Sanfey et al., 2003; Yu et al., 2015). P3 is influenced by the valence and magnitude of an outcome. Positive feedback and greater rewards will elicit greater P3 (Bellebaum et al., 2010). In the UG task, fair offers were more in line with fairness norms and benefits individuals more than unfair offers, inducing a larger P3 amplitude (Wu et al., 2011; Wu & Zhou, 2009). However, in the research of Liu et al. (2018, 2020), it was found that P3 did not be modulated by empathy, and the valence effects of P3 only appeared in self-condition, indicating that P3 might only be associated with self-relevant information processing.

### 1.4 Research contents and hypothesis

In the present study, we used UG paradigm and ERP technology to simulate a real interpersonal interaction situation to examine the influence of empathic concern on fairness-related decision making and the time-dynamic characteristics underlying fairness processing. Based on the empathy-altruism hypothesis, people would experience strong empathy for people in need or in distress (Batson & Ahmad, 2001; Batson & Moran, 1999). Therefore, in our study, the proposers in the empathy condition were left-behind students from a school in a remote, impoverished region (in need), whereas the proposers in the non-empathy condition were ordinary students in a normal urban school (not in need). This manipulation drew on the studies of Liu et al. (2018, 2020) and its effectiveness of manipulation has been confirmed. The participants, as responders, were asked to decide whether to accept the fair, advantageous unfair and disadvantageous unfair offers or not. We hypothesized that there would be differences in acceptance rate and subjective ratings between empathy and non-empathy context for different offers. Besides, as AN1 was modulated by the interactive object (Wang et al, 2014), and empathy was induced by manipulating the identity of proposers in the present study, we hypothesized that empathy would modulate the AN1 in different fairness conditions. Meanwhile, empathy could induce individuals' altruistic motivation, so that the motivation would be different between empathy and non-empathy condition, adjusting the P2 amplitude. In addition, it has been shown in previous studies that MFN was moderated by empathy and fairness, and considering that empathy could affect individuals' attention to outcomes, we hypothesized that MFN induced by different fairness conditions would be different in empathy context, but there would be no difference in non-empathy context. Finally, we still hypothesized that P3 would not be modulated by empathy, but only exhibit fairness effect, which was consistent with previous studies (Liu et al., 2018, 2020).

## 2 Method

### 2.1 Participants

The participants were 42 non-economics and non-psychology undergraduates, all of whom were right-handed, in good health, without neurological or psychiatric disorders history. They had normal or corrected-to-normal vision. All the participants volunteered to participate in the experiment and signed consent form. Five participants were excluded due to an insufficient number of trials after artifacts were removed. The final sample for analysis consisted of the data of 37 participants (26 females, 18~24 years old, $M \pm SD = 21.00 \pm 2.07$ years old). Sample size estimation was performed using G*Power v.3.1 (Faul et al., 2009). According to the analysis ($f = 0.25$, $\alpha = 0.05$, $\beta = 0.95$, ANOVA: repeated measures, within-subject factors), a total sample size of 28 participants was required to detect reliable effects, and the number of participants in our study met the requirements.

### 2.2 Procedure

First of all, participants clarified that they need to decide whether to accept the proposers' allocation proposal for 10 yuan. They were informed that the proposers were left-behind students from a school in a remote, impoverished region and ordinary students from a normal urban school. The family status and school infrastructure of left-behind students are relatively poor compared with those of ordinary students. The participants were presented with textual materials and pictures describing the family status, school infrastructure, teachers and enrollment rate of the left-behind students and ordinary students to induce state empathic concern (Liu et al., 2018, 2020). In order to test the validity of empathy manipulation adopted in our study, we recruited another 22 participants. After the same empathy manipulation steps, participants were asked to fill out the Empathy Response Questionnaire (ERQ) for manipulation check (Batson et al., 1997, 2002). A 7-point scale was used to measure the extent to which the participants felt compassionate/concerned/touched (Pfattheicher et al., 2019). The three items were of high internal consistency reliability (Cronbach's $\alpha = 0.88$), so that the state empathy score was obtained by averaging the three scores. A one-way repeated measures ANOVA of state empathy scores in both contexts revealed that the scores in the left-behind student context were significantly higher than those in the ordinary student context ($5.56 \pm 0.95$ vs. $2.39 \pm 1.35$, $F$ (1, 21) = 129.68, $p < 0.001$, $\eta_p^2 = 0.86$), indicating that the left-behind students' context successfully elicited state empathic concern in individuals.

In order to increase the credibility of the experiment, we emphasized to the participants that each offer was put forward by different proposers. Since the game could not be played at the same time and place, all the offers were collected in advance. We also informed participants that we would count and distribute the benefits between them and the proposers according to their decisions, so as to ensure participants believed that

their choices would indeed affect the interests of others. The proposers' reward depended on the decisions of participants. If participants accepted the offer, the proposer would get the corresponding amount of money. Otherwise, the proposer would get nothing. The payment of participants consisted of three parts: 15 rounds were randomly selected from offers participants chose to accept in empathy and non-empathy context, and basic remuneration for the participation. This payment structure implied that if participants wanted to get as much as possible, they would need to reject both fair offers and disadvantageous unfair offers and only accept advantageous unfair offers. Unlike previous studies, this manipulation could distinguish whether participants choose to accept disadvantageous unfair offers out of self-interest or purely altruistic motivations. Meanwhile, it could also examine whether individuals would give up self-interest maximization to pursue fairness through whether individuals accepted fair offers or not. Generally speaking, the manipulation provided a more comprehensive examination of participants' behavioral motivations. The payments were set to make participants believe that their decisions were tied to their true experimental rewards, thus ensuring that participants would make more realistic decisions. In fact, the participants were paid 75 yuan per person, which was compensation of their time. However, participants did not know that each of them was paid the same.

### 2.3 Design and materials

The experiment used a 2 (state empathic concern: empathy vs. non-empathy) × 3 (fairness: fair vs. disadvantageous unfair vs. advantageous unfair) within-subject design. A total of 10 proposals were set up: fair condition (4/6, 5/5, 6/4, 25 trials each); disadvantageous unfair condition (8/2, 9/1, 30 trials each); advantageous unfair condition (2/8, 1/9, 30 trials each); filling condition (3/7, 7/3, 30 trials each). The number in front of the diagonal line indicated the amount of money that the proposers gave themselves. The setting of fairness conditions was consistent with the study of Wu and Zhou (2012). The dependent variables were acceptance rate, subjective ratings and ERP components. The entire task was divided into two tasks: empathy and non-empathy. Each task was divided into five blocks of 51 trials each with a brief break between blocks. The order of the two tasks was balanced between the participants.

Participants were also required to rate their emotional reactions and fairness perception to the currently received offer. The emotional reaction rating was a 9-point scale (1 = extremely negative and 9 = extremely positive) (Bradley et al., 2001; Hewig et al., 2011; Hu & Mai, 2021). The fairness perception was a 7-point scale (1 = extremely unfair and 7 = extremely fair) (Leliveld et al., 2012; Pillutla & Murnighan, 1996).

### 2.4 Research flow

Stimulus presentation and behavioral data acquisition were conducted using E-Prime 3.0 software. The procedure is shown in Figure 1. In each block, the identity of the proposer would be displayed on the screen. For each single trial, a fixation cross was presented for 800~1200 ms, followed by a blank screen of 500 ms. Later, the offer would be presented for 1200ms. The above number indicates the amount of the proposer. Then a blank screen would be presented for 500ms again. After that, participants should decide to accept the offer by pressing F or reject it by pressing J. In each block, each type of proposal would randomly have once emotion rating and once fairness perception rating.

### 2.5 EEG recording and preprocessing

EEG data were recorded using Biosemi EEG recording system with a 32-conductor electrode cap. The signal was amplified (0.1~100 Hz band pass), digitized at a sampling rate of 512 Hz and stored for off-line analyses. Vertical and horizontal electrooculograms (EOGs) were collected with electrodes placed on the left supraorbital and infraorbital and on the outer canthi of the left and right eyes, respectively. All interelectrode impedance was kept below 5 kΩ during recording. The electrode value of the left mastoid was used as a reference when EEG was recorded, and the average value of the bilateral mastoid was used for re-reference during off-line analysis. The EEG data were processed offline using eeglab toolbox (Version 12.0.2.6 b) of Matlab (2013a) software. The EEG data was segmented into epochs from 200 ms before the onset of the stimuli to 800 ms after the stimuli. Then, the ocular artifacts were corrected by using sentence (Gratton et al., 1983). A 30 Hz low-pass filter was applied to the EEG. Epochs containing artifacts exceeding ± 75 μV were excluded. Finally, epochs were averaged for different conditions.

According to the characteristics of the grand-average ERP waveform, we extracted AN1 (minimum peak value of 100~200 ms), P2 (mean value of 230~280 ms), MFN (mean value of

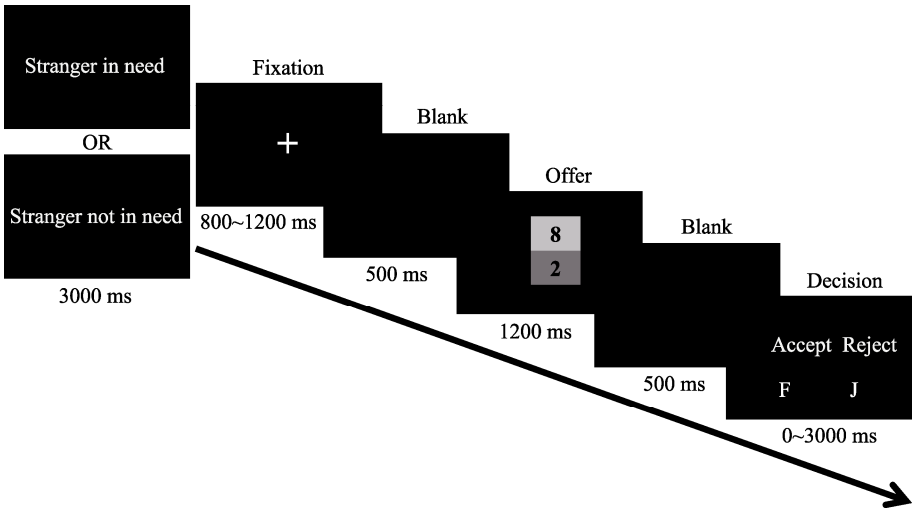



*Figure 1.* Illustration of a single trial of the multi-round one-shot UG task.

300~450 ms) and P3 (mean value of 250~350 ms). According to the topographic distribution of ERP components and previous studies (Yeung & Sanfey, 2004), AN1, P2, MFN and P3 were calculated through 9 electrode points (F3, Fz, F4, C3, Cz, C4, P3, Pz, P4). The results showed that the amplitude of AN1, P2 and MFN were maximum at Fz site, and the amplitude of P3 were maximum at Pz site. Therefore, we selected Fz and Pz as the electrodes for further statistical analysis.

### 2.6 Temporal PCA

Since P2, MFN and P3 might overlap in ERP waveforms, we used principal component analysis (PCA) to decompose the ERP components (Foti et al., 2011). We used Evoked ERP/EPO toolkit based on Matlab to perform the temporal PCA (Zhang et al., 2020). A temporal PCA with Promax rotation was first performed at all time points of the averaged ERPs of each participant, and then the components of interest were selected and projected to the electrode field based on the following three factors: the polarity and latency of temporal components, the polarity and location of the activated regions of spatial components and similarity of topographies of the components of interest in each condition. Finally, P2 (mean value of 200~225 ms at Fz), MFN (mean value of 360~400 ms at Fz) and P3 (mean value of 460~510 ms at Pz) were selected for statistical analysis.

To confirm the identification of PCA factors, we calculated the Pearson correlations between the PCA factors and the corresponding ERP components in conventional time domain analysis. Results showed that P2 and PCA-P2 were highly correlated ($r = 0.84$, $p < 0.001$), and MFN MFN was highly correlated with PCA-MFN ($r = 0.99$, $p < 0.001$). However, considering that the latency of P3 and PCA-P3 obtained in conventional time domain analysis were not consistent, and the temporal and spatial domain characteristics of PCA-P3 were more in line with the component characteristics of interest, we mainly focused on PCA-P3 results and did not further examine the correlation between them.

### 2.7 Statistical analysis

All data analyses were performed in SPSS 25.0 software. Behavior data, EEG data and PCA factors were analyzed using 2 (state empathic concern: empathy vs. non-empathy) × 3 (fairness: fair vs. disadvantageous unfair vs. advantageous unfair) within-subject repeated measures ANOVA. When the assumption of sphericity did not hold, the degree of freedom of F value ratio were corrected by Greenhouse-Geisser. Bonferroni corrections were performed for post hoc testing of significant main effects. Simple effect analyses were performed to test significant interactions. Descriptive statistics were expressed as mean ± standard deviation ($M \pm SD$). The significance level of all statistical analysis was set to 0.05, and the effect size of all variance analysis was reported by partial eta-squared ($\eta_p^2$), with 0.01 representing a small effect size, 0.06 representing a medium effect size, and 0.14 representing a large effect size (Hu & Dai, 2011).

## 3 Results

### 3.1 Behavioral results

#### 3.1.1 Acceptance rate

A significant main effect of empathy was confirmed, $F(1, 36) = 4.87$, $p = 0.034$, $\eta_p^2 = 0.12$, with the acceptance rate of empathy condition higher than that of non-empathy condition (0.67 ± 0.21 vs. 0.62 ± 0.16). The main effect of fairness was also significant, $F(2, 72) = 90.89$, $p < 0.001$, $\eta_p^2 = 0.72$, and Bonferroni corrections post hoc tests indicated higher acceptance rates for fair offers (0.84 ± 0.26) and advantageous unfair offers (0.88 ± 0.23) than disadvantageous unfair offers (0.20 ± 0.29, $p$s < 0.001), with no significant difference between fair and advantageous unfair offers ($p > 0.1$). The interaction effect of empathy × fairness was significant, $F(1.64, 59.12) = 14.04$, $p < 0.001$, $\eta_p^2 = 0.28$. Simple effects analysis showed that the acceptance rate of empathy condition was higher than that of non-empathy for disadvantageous unfair offers (0.30 ± 0.40 vs. 0.11 ± 0.25, $p < 0.001$), while for advantageous unfair offers, the acceptance rate of empathy condition was lower than that of non-empathy condition (0.84 ± 0.31 vs. 0.92 ± 0.20, $p = 0.025$). Besides, there was no significant difference between the two conditions for fair offers ($p > 0.1$). (See Figure 2A.)

### 3.2 Subjective ratings

#### 3.2.1 Emotion rating

The main effect of fairness was significant, $F(1.34, 47.69) = 33.23$, $p < 0.001$, $\eta_p^2 = 0.48$, participants' emotional reactions were more positive for fair offers (5.61 ± 0.72) and advantageous unfair offers (6.02 ± 1.55) than disadvantageous unfair

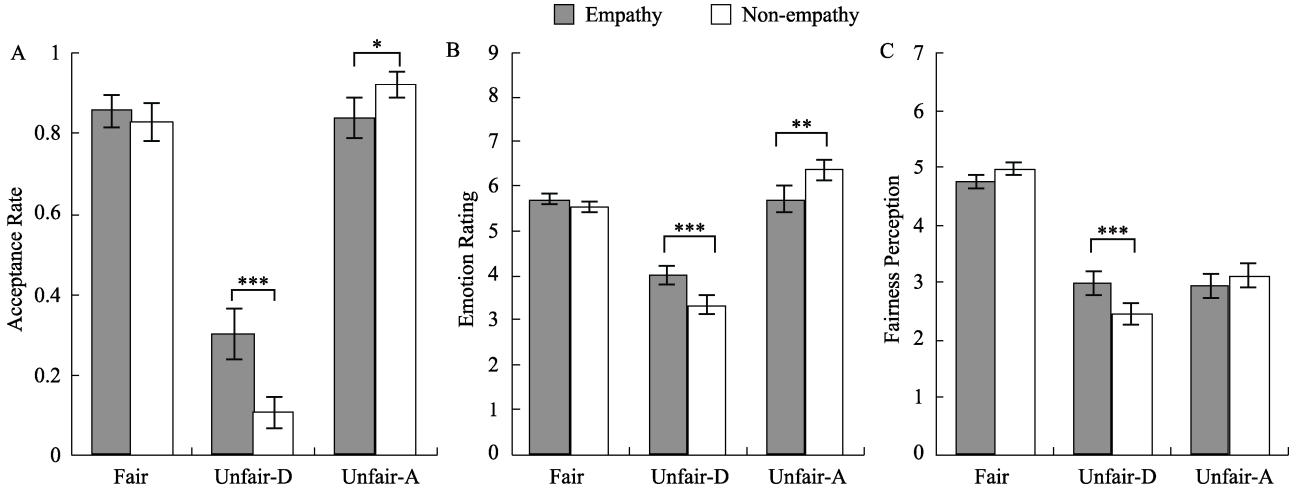



*Figure 2.* Histogram of interaction effect between empathy and fairness on ARs (A), emotion ratings (range from 1~9) (B) and fairness perception (range from 1~7) (C). Unfair-D means disadvantageous unfair. Unfair-A means advantageous unfair.

*Note*. Error bars indicate standard error of the mean (SEM); $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

offers (3.67 ± 1.17, *p*s < 0.001). There was no significant difference between fair and advantageous unfair offers ($p = 0.471$). The interaction effect of empathy × fairness was significant, $F(2, 72) = 13.95$, $p < 0.001$, $\eta_p^2 = 0.28$, with more negative reactions in non-empathy condition than in empathy condition to disadvantageous unfair offers (3.32 ± 1.23 vs. 4.02 ± 1.35, $p < 0.001$). For advantageous unfair offers, participants experienced more positive reactions in non-empathy condition than in empathy condition (6.35 ± 1.55 vs. 5.68 ± 1.80, $p = 0.002$), with no significant difference for fair offers ($p = 0.214$). (See Figure 2B.) The main effect of empathy was not significant ($p = 0.509$).

### 3.2.2 Fairness perception

The main effect of fairness was significant, $F(2, 72) = 58.64$, $p < 0.001$, $\eta_p^2 = 0.62$, with higher fairness perception for fair offers (4.88 ± 0.60) than disadvantageous unfair (2.72 ± 1.08) and advantageous unfair offers (3.03 ± 1.25, *p*s < 0.001), while fairness perception for disadvantageous and advantageous unfair offers were not significantly different ($p = 0.528$). The interaction effect of empathy × fairness was significant, $F(1.70, 61.26) = 13.27$, $p < 0.001$, $\eta_p^2 = 0.27$. Fairness perception were lower in non-empathy condition than that in empathy condition (2.44 ± 1.08 vs. 2.99 ± 1.23, $p < 0.001$) for disadvantageous unfair offers, whereas there was no significant difference between empathy and non-empathy condition for fair and advantageous unfair offers ($p = 0.057$; $p = 0.083$). (See Figure 2C.) The main effect of empathy was not significant ($p = 0.445$).

## 3.3 ERP results−conventionally averaged ERPs

### 3.3.1 AN1

The interaction effect of empathy × fairness was significant, $F(2, 72) = 3.83$, $p = 0.026$, $\eta_p^2 = 0.10$, with the AN1 amplitude induced by non-empathy condition (−4.67 ± 2.55 μV) being more negative than that of empathy condition (−3.90 ± 2.05 μV, $p = 0.030$) in advantageous unfair condition. However, there was no significant difference between empathy and non-empathy condition for both fair and disadvantageous unfair offers. (See Figure 3.) The main effects of both empathy and fairness were not significant (*p*s > 0.1).

### 3.3.2 P2

The interaction between empathy and fairness was significant, $F(2, 72) = 4.88$, $p = 0.010$, $\eta_p^2 = 0.12$. For advantageous unfair offers, the P2 amplitude was greater of empathy condition (−1.10 ± 3.13 μV) than that of non-empathy condition (−1.84 ± 3.28 μV, $p = 0.049$). For disadvantageous unfair offers, the P2 amplitude was slightly larger in non-empathy condition (−1.08 ± 3.28 μV) than in empathy condition (−1.96 ± 2.78 μV, $p = 0.096$); the difference between the two was not significant for fair offers ($p = 0.726$) (See Figure 3). The main effects of empathy and fairness were not significant (*p*s > 0.1).

### 3.3.3 MFN

The interaction effect of empathy × fairness was significant, $F(2, 72) = 3.97$, $p = 0.023$, $\eta_p^2 = 0.10$. As shown in Figure 4, in empathy context, the MFN amplitude induced by disadvantageous unfair offers (−4.02 ± 2.98 μV) was significantly lower than that of advantageous unfair offers (−2.79 ± 3.33 μV, $p = 0.008$), slightly lower than that of fair offers (−3.20 ± 3.26 μV, $p = 0.081$). There was no significant difference between fair and advantageous unfair offers ($p = 0.715$). In non-empathy context, there were no significant differences among the three (*p*s > 0.1). The main effects of both empathy and fairness were not significant (*p*s > 0.1).

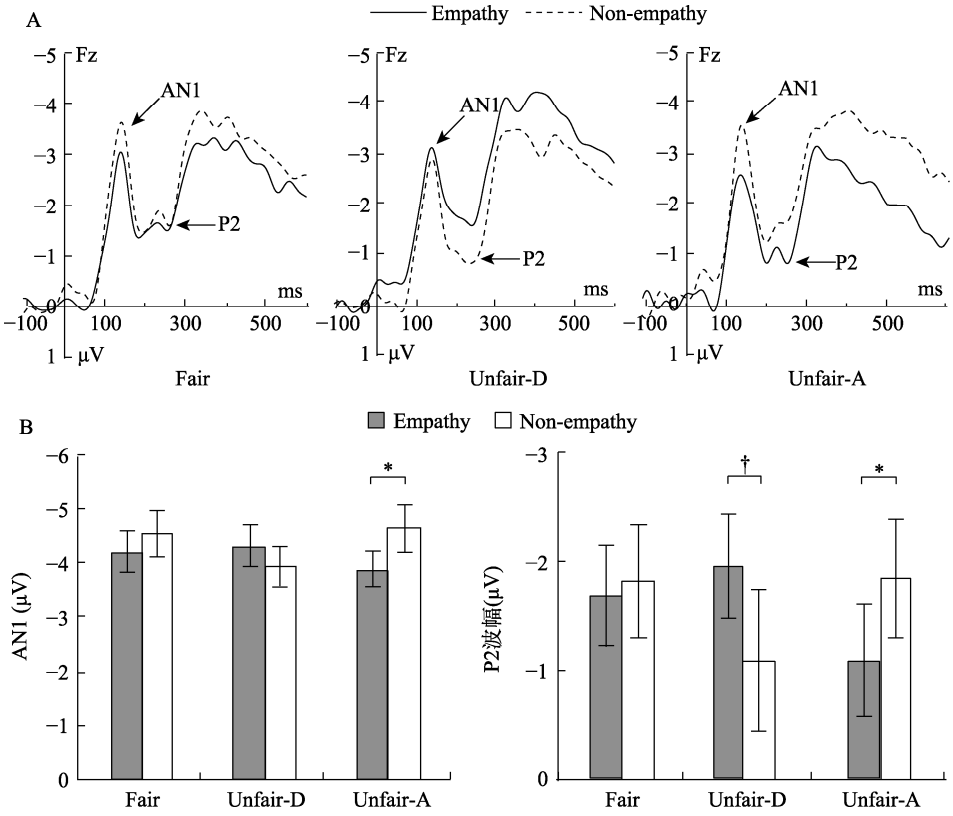



*Figure 3.* (A)The Grand-average ERP waveforms from Fz electrode sites for the 3 fairness conditions, respectively; (B) Histogram of interaction effect between empathy and fairness on AN1 and P2. Unfair-D means disadvantageous unfair. Unfair-A means advantageous unfair.
*Note*. Error bars indicate standard error of the mean (SEM); $^{\dagger}p < 0.1$, $^{*}p < 0.01$.

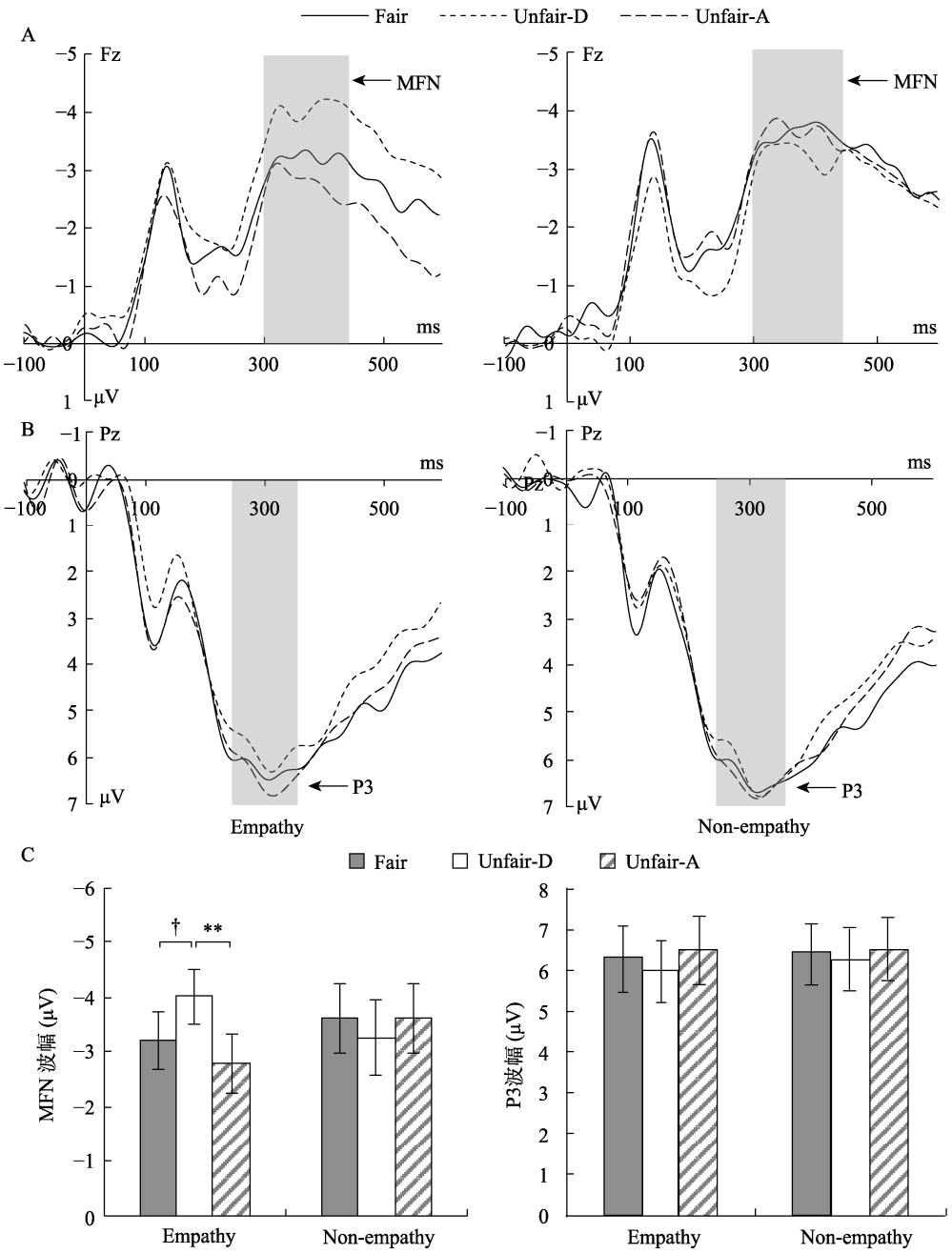



*Figure 4*. The Grand-average ERP waveforms from Fz(A) and Pz(B) sites for empathy and non-empathy condition, respectively; (C) Histogram of interaction effect between empathy and fairness on MFN and P3.Unfair-D means disadvantageous unfair. Unfair-A means advantageous unfair.

*Note*. Error bars indicate standard error of the mean (SEM); $^{\dagger}p < 0.1$, $^{**}p < 0.01$.

#### 3.3.4 P3

The main effects and interaction effect of empathy and fairness were not significant (*p*s > 0.1).

### 3.4 ERP result: temporal PCA

#### 3.4.1 PCA-P2

The interaction effect of empathy × fairness was significant, $F(2, 72) = 3.80$, $p = 0.027$, $\eta_p^2 = 0.10$. In the empathy context, advantageous unfair offers (−1.30 ± 2.59 μV) elicited slightly larger P2 than fair offers (−1.84 ± 2.70 μV, $p = 0.066$), and there was no significant difference for disadvantageous unfair offers (−1.99 ± 2.87 μV, $p = 0.101$). (See Figure 5A.) In the non-empathy context, there were no significant differences among the three (*p*s > 0.1). The main effects of empathy and fairness were not significant (*p*s > 0.1).

#### 3.4.2 PCA-MFN

The interaction effect of empathy × fairness was significant, $F(2, 72) = 4.60$, $p = 0.013$, $\eta_p^2 = 0.11$. In the empathy context, the disadvantageous unfair condition (−5.47 ± 3.57 μV) elicited a more negative-going MFN than the advantageous unfair condition (−3.88 ± 4.15 μV, $p = 0.007$), which was slightly more negative than that of fair condition (−4.52 ± 3.99 μV, $p = 0.090$). There was no significant difference between advantageous unfair and fair condition ($p = 0.478$). In the non-empathy context, there were no significant differences among the three (*p*s > 0.1). (See Figure 5B.) None of the main effects of empathy and fairness were significant (*p*s > 0.1).

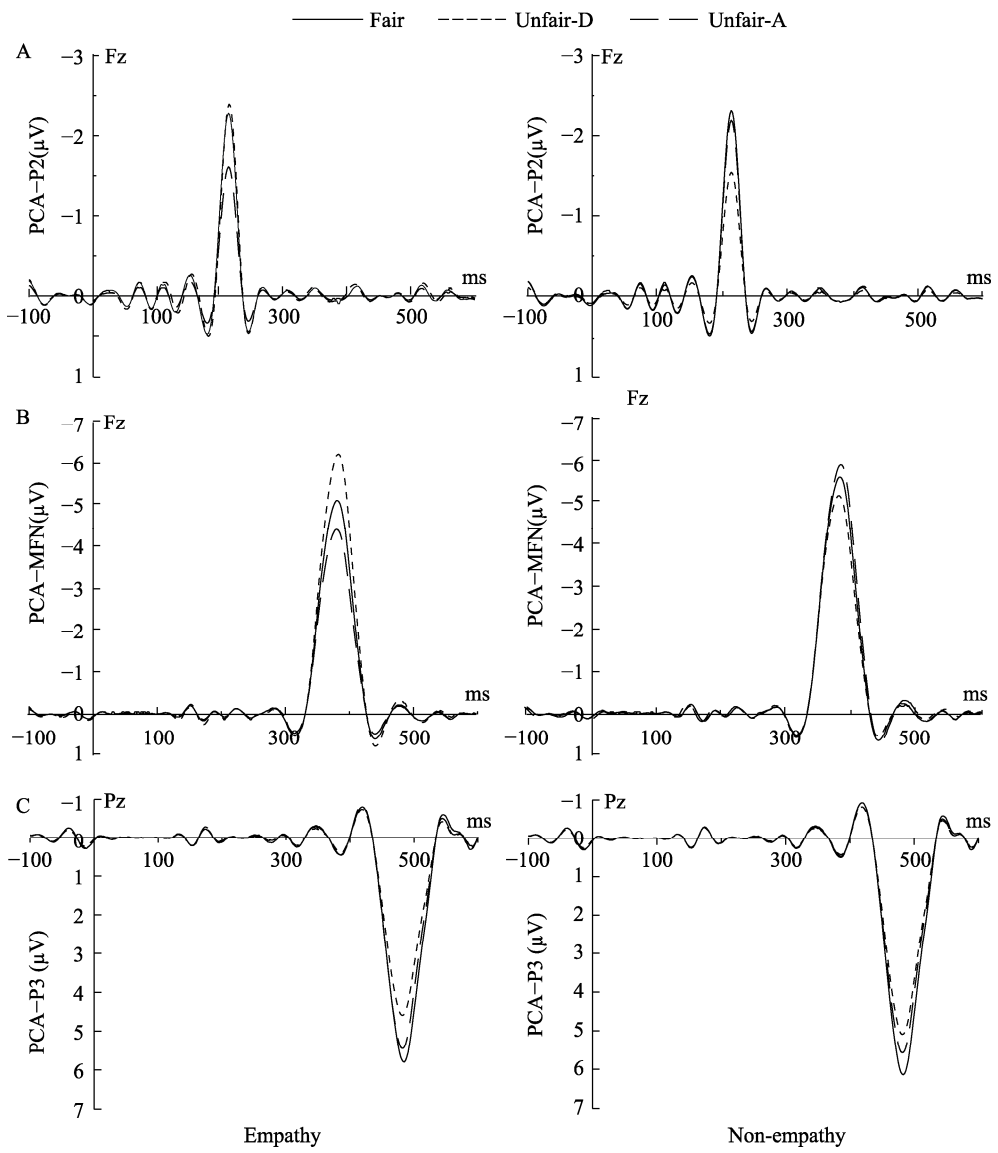



*Figure 5*. Grand-average waveforms of PCA-P2 (A) and PCA-MFN (B) at the Fz electrode site and PCA-P3 (C) at the Pz electrode site. Unfair-D means disadvantageous unfair. Unfair-A means advantageous unfair.

#### 3.4.3 PCA-P3

The main effect of fairness was significant, $F(2, 72) = 5.73$, $p = 0.005$, $\eta_p^2 = 0.14$. The amplitude of P3 of fair condition was significantly greater than that of disadvantageous unfair condition (4.86 ± 4.30 μV vs. 3.92 ± 4.33 μV, $p = 0.004$), but was not significantly different from that of advantageous unfair condition (4.46 ± 4.47 μV, $p = 0.412$). P3 for disadvantageous unfair offers and advantageous unfair offers were not significantly different ($p = 0.238$). (See Figure 5C.) Neither the main effect of empathy nor the interaction effect of empathy × fairness reached significant ($ps > 0.1$).

## 4 Discussion

This study explored the modulating effect of empathic concern on fairness-related decision making and its temporal characteristics of neural activities using UG paradigm and ERP technology. The results demonstrated that empathic concern and fairness interacted with each other to influence individuals' fairness-related decision making behaviors and subjective ratings, with higher acceptance rate, emotional rating and fairness perception rating in empathy condition than in non-empathy condition for disadvantageous unfair offers, while the opposite results were found for advantageous unfair offers. The EEG results revealed that empathy condition elicited more negative-going AN1 than non-empathy condition for advantageous unfair offers, while the difference between empathy and non-empathy condition for disadvantageous unfair and fair offers was not significant, suggesting that the empathic concern can influence early attentional process of fairness-related decision making. In addition, compared to non-empathy condition, advantageous unfair offers in empathy condition induced larger P2 amplitude, while an opposite trend was observed in disadvantageous unfair condition, reflecting the effect of empathic concern on motivation in fairness processing. In empathy condition, disadvantageous unfair offers elicited a more negative-going MFN, while there were no significant differences in non-empathy condition, reflecting the cognitive and emotional processing modulated by empathic concern. We also used PCA to separate potentially overlapping ERP components and found that PCA-P2 and PCA-MFN showed similar patterns to conventional ERP results.

Notably, PCA-P3 was only influenced by fairness, but not modulated by empathy.

### 4.1 Empathy modulates fairness perception, emotional reactions and decision-making behavior

Both emotional reactions and fairness perception are important factors that motivate individuals to engage in irrational behaviors that violate the principle of self-interests maximization (such as rejecting unfair offers) (Fehr & Gachter, 2002; Matarazzo et al., 2016; Sanfey et al., 2003). In the present study, individuals' emotional reactions, fairness perception and decision-making behaviors were all modulated by empathy. The fairness perception, emotion rating and acceptance rate in empathy condition were higher than those in non-empathy condition, reflecting that empathy could increase individuals' tolerance for disadvantageous unfair offers. Whereas advantageous unfair offers were contrary to individuals' altruistic motivation, emotional reactions of empathy condition were more negative than non-empathy, and the acceptance rate decreases accordingly. It also reflects that the motivation in empathy condition comes more from altruistic motivation, suggesting that when empathy conflicts with fairness norms, individuals would give priority to guiding behaviors through empathy. Moreover, based on the manipulation of payment in this study, the benefit-maximizing choice was to accept only advantageous unfair offers, while the acceptance rate of fair offers was actually as high as advantageous unfair offers, which reflects the motivation to pursue fairness rather than just to maximize self-interest.

### 4.2 Empathy modulates the early automation stage of fairness processing

The decision-making process involves reasoning processing of theory of mind, such as inferring and forming expectations of other's goals and strategies (Polezzi et al., 2008; Wu & Zhou, 2012). AN1 is considered to be associated with stronger anticipatory activities (Boudreau et al., 2009; Bouwer et al., 2020), and the experimental context and interactive objects will affect the attentional bias and expectations of participants (Wang et al., 2014; Wu et al., 2011). Therefore, modulated by the level of empathy, participants might reason differently about the offers of different proposers and form decision expectations in advance. People are more inclined to help those who are willing to help others. In empathy context, when people want to help a stranger, they usually assume that the other party is worthy of their help. Thus, they tend to think that the other party is more cooperative and reciprocal, so that fair and advantageous unfair offers is in line with expectations. In contrast, proposers in non-empathy context might be more inclined to pursue self-interests maximization, so advantageous unfair offers violate the expectations of participants. Therefore, advantageous unfair offers were new stimulus for participants so that enhanced visual attention and induced more negative-going AN1 (Kubota & Ito, 2007). According to this interpretation, disadvantageous unfair offers in empathy condition should induce more negative-going AN1 than that in non-empathy condition. There was a numerical trend, but it was not statistically significant.

Empathy also modulated P2 that represented motivational meaning: in terms of advantageous unfair offers, participants only pay attention to their own interests in non-empathy context, while in empathy context, participants pay attention to both their own interests and the others' interests. Moreover, the unequal distributions of self and others violated their altruistic and equity-seeking motivations. Their motivational salience was stronger, extracted and characterized in P2 and elicited a larger P2 amplitude (Carretie et al., 2004; Luck & Hillyard, 1994). On the contrary, in the empathy context, choosing to accept disadvantageous unfair offers would violate the principle of maximizing self-interests. However, choosing to reject would harm the interests of the other party and conflict with altruistic motivation. Therefore, it was a situation of conflicting interests for participants. However, in the non-empathy context, participants only paid attention to their own interests and rejecting the offers would not affect their own benefits. It was a consistent situation of interests, thus inducing a larger P2 (Boudreau et al., 2009).

In conclusion, empathy modulated the early automatic processing stage of fairness-related decision making, modulated early attention represented by AN1 by influencing participants' expectations, and modulated P2 amplitude by influencing their behavioral motivations.

### 4.3 Empathy modulates the mid-term semi-automated stage of fairness processing

Most studies suggest that MFN may reflect social pain, violation of social norms, cognitive conflict and emotion conflict, or worse than expected outcomes (Boksem & de Cremer, 2010; Polezzi et al., 2008; Spape et al., 2019; Van der Veen & Sahibdin, 2011). In the present study, the fairness effect of MFN was only found in the empathy condition, but not the in non-empathy condition. The possible explanation for this is that the setting of participants' payment in this study is different from previous studies. That is, rejecting unfair offers would not affect their own benefits. Therefore, participants could only pay attention to their own interests, need not to care about others' gains, and the cognitive processing resources allocated to fairness were reduced, so that MFN did not show any differences (Wu & Zhou, 2012). In the empathy condition, participants were concerned with others' gains, so they would process fairness conditions and show differences. Although MFN did not show any differences in the non-empathy condition, there was no significant difference of MFN amplitude between empathy condition due to the consumption of cognitive resources. Therefore, it is assumed that MFN also reflected the input and consumption of cognitive resources in the decision-making process (Polezzi et al., 2008).

Considering the higher altruistic motivation in empathy condition, participants were willing to sacrifice their own interests to protect each other's welfare, and correspondingly, they expected those they were trying to help to be more reciprocal and cooperative. Thus, the larger MFN for disadvantageous unfair offers reflected the violation of expectations and negative emotions (e.g., disappointment, betrayal, etc.) (Moser et al., 2014). In any case, it is clear that empathy does modulate the rapid and semi-automatic processing in the middle stage fairness-related decision making.

### 4.4 Research limitations and prospects

There are several limitations in our study. First, we should provide a uniform and standardized description of the background information of the "left-behind children" and "ordinary children" in the empathic concern manipulation to avoid individual differences in understanding these two groups. In addition, the number of male and female participants in our study differed significantly. Although no effect of gender was found in the statistical tests, the number of genders should be balanced in future studies. Finally, relevant trait variables such as

trait empathic concern and fairness preferences were not measured in our study.

## 5 Conclusion

Our study investigated the influence of empathic concern on fairness-related decision making behavior and revealed the temporal characteristics of neural activities underlying it through ERP technology. Our findings revealed that empathic concern interfered with individuals' fairness-related decision making. Individuals' tolerance for unfair behaviors increased when they showed empathic concern for others, indicating that when empathy conflicted with fairness norms, individuals would preferentially process empathy to guide behavior. The time course of neural activities for this effect was that empathic concern modulated the early automatic and middle semi-automatic stages of fairness processing, suggesting that empathy would influence individuals' early attention allocation and behavioral motivation, as well as negative emotional reactions and cognitive resources. The series of findings provides a new way to explain individuals' fairness-related decision making behavior in particular contexts, suggesting that we can influence individuals' fairness perception and behavior by manipulating the level of empathy. Meanwhile, it also helps us to further understand how empathic concern modulate fairness processing.

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